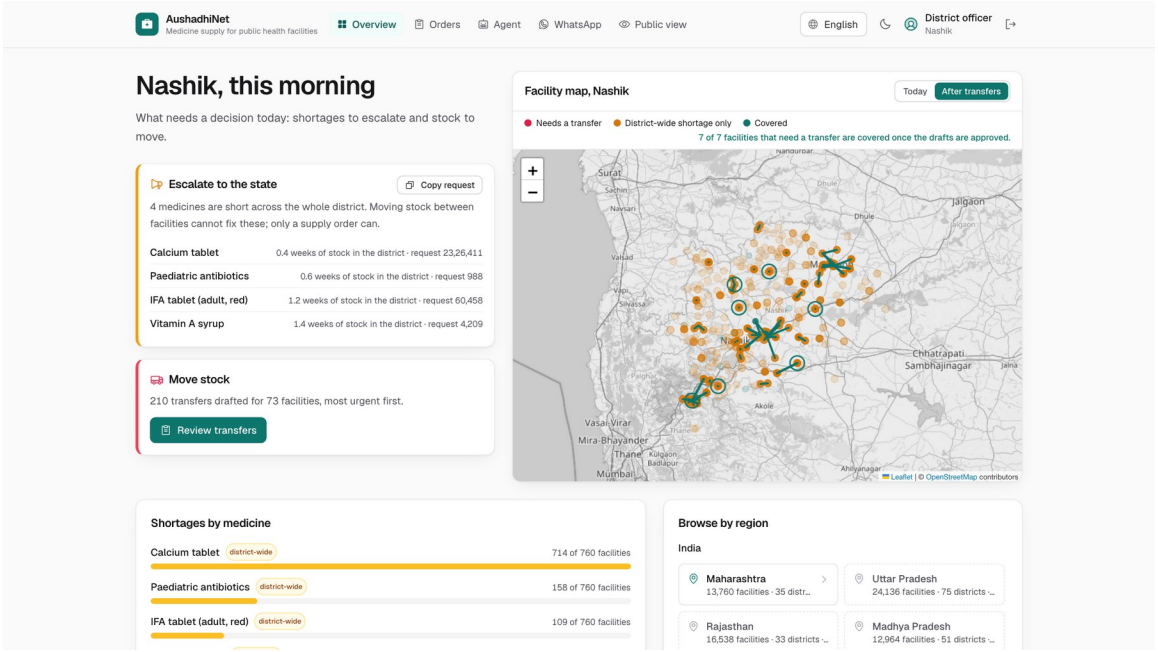
Build with AI: Code for Communities · Problem Statement 03

AushadhiNet

Early warning and rebalancing for public-health medicine supply, reported over WhatsApp.



Live on Google Cloud Run: aushadhinet-frontend-449585498230.asia-south1.run.app

Code: github.com/akshat333-debug/AushadhiNet

The problem

Clinics run dry while neighbours hold spare stock

Primary health centres run out of ORS, zinc, iron tablets and antibiotics.

Stock is reported late, on paper, and rarely compared across facilities.

Shortage and surplus sit a few kilometres apart; nobody sees both in time.

Frontline nurses won't adopt yet another app.

Who it serves

Built around the district officer's morning

District officer

District health officer or pharmacist, on a laptop: what to escalate to the state, which transfers to approve, most urgent first.

Nurse at the facility

ANM, staff nurse or pharmacist at a PHC or sub-centre, on a basic Android phone: reports in 30 seconds on WhatsApp, in their own language.

Public and planners

Citizens see district shortage totals, never a facility name. State planners see where to roll out next.

The nurse never opens the website: WhatsApp is the interface. The officer gets decisions, not dashboards.

How it works

Report → Forecast → Rebalance → Human approves

1 Report

Nurse sends stock on WhatsApp by text, voice note or register photo, in 7 languages; every machine read is confirmed.

2 Forecast

Next-week demand and stock-out probability per facility and medicine; model chosen by measurement on real HMIS data.

3 Rebalance or escalate

OR-Tools drafts transfers from nearby surplus. Medicines short district-wide go to the state as a supply request.

4 Approve

Officer approves; signed order to both facilities. The AI agent explains and drafts, never approves.

Nurse channel

WhatsApp, 7 languages, no new app

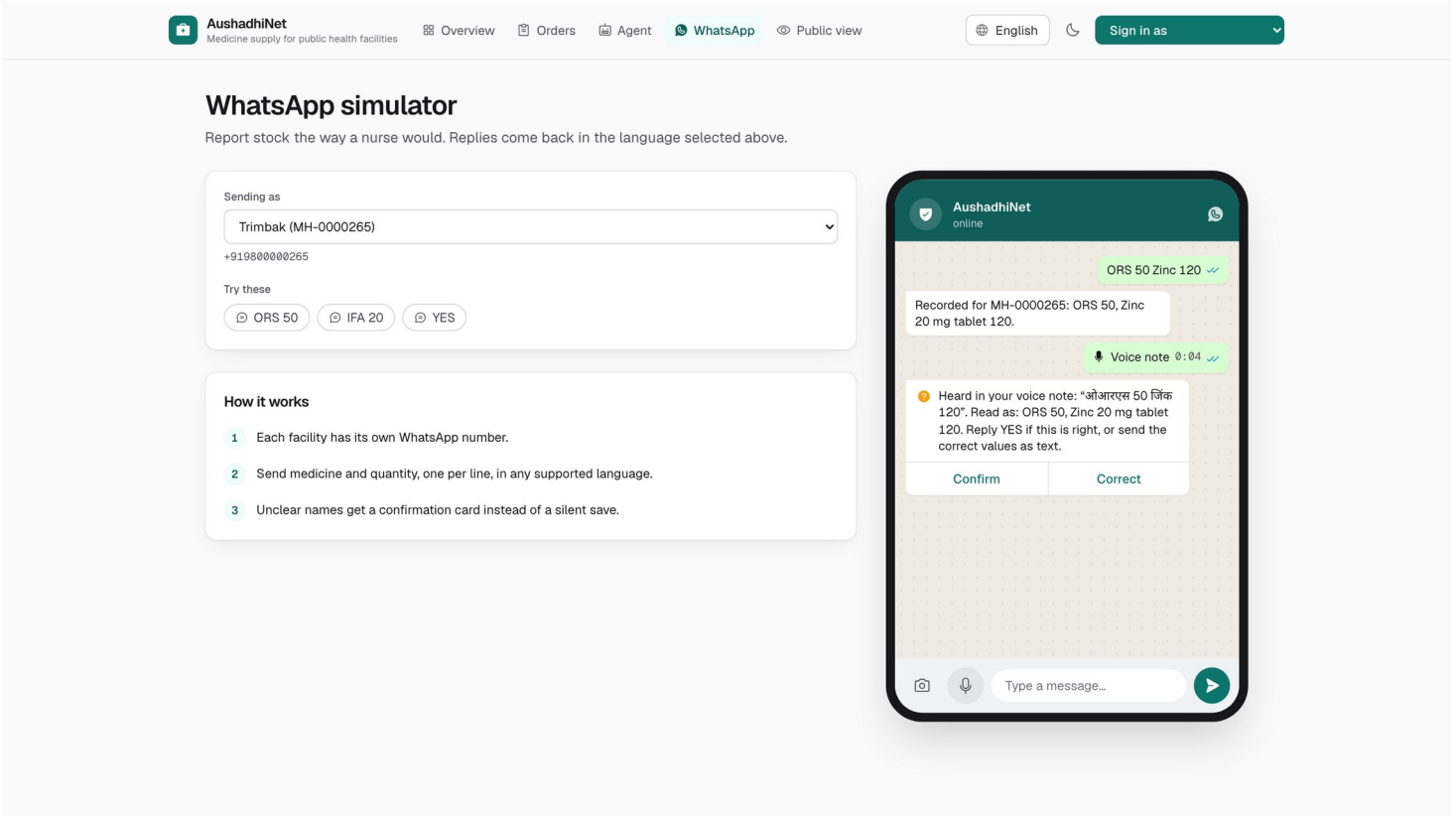
Text, a voice note or a photo of the register

Chirp 2 hears the note and detects the language; Gemini reads every line of a photo

Several items in one message: “ORS 50 Zinc 120”

Nothing is saved until the nurse replies YES

English, Hindi, Marathi, Bengali, Tamil, Telugu, Kannada



Officer view

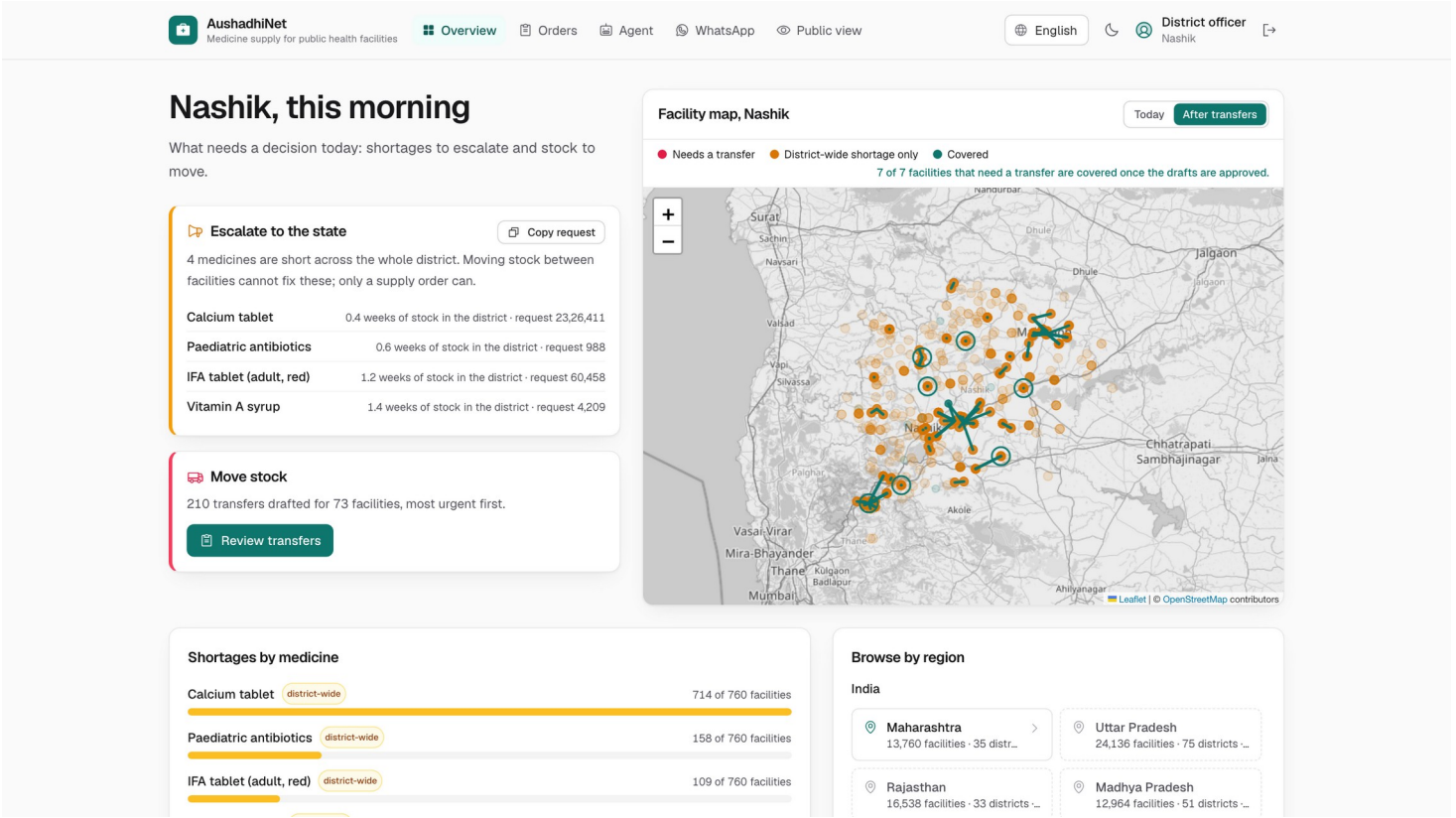
The morning briefing: escalate or move

Short across the district: a supply request to the state, with quantities

Local imbalance: OR-Tools drafts transfers, most urgent first

Map before and after the drafted transfers

Officer approves; Cloud KMS signs; both facilities alerted on WhatsApp



Guardrailed agent

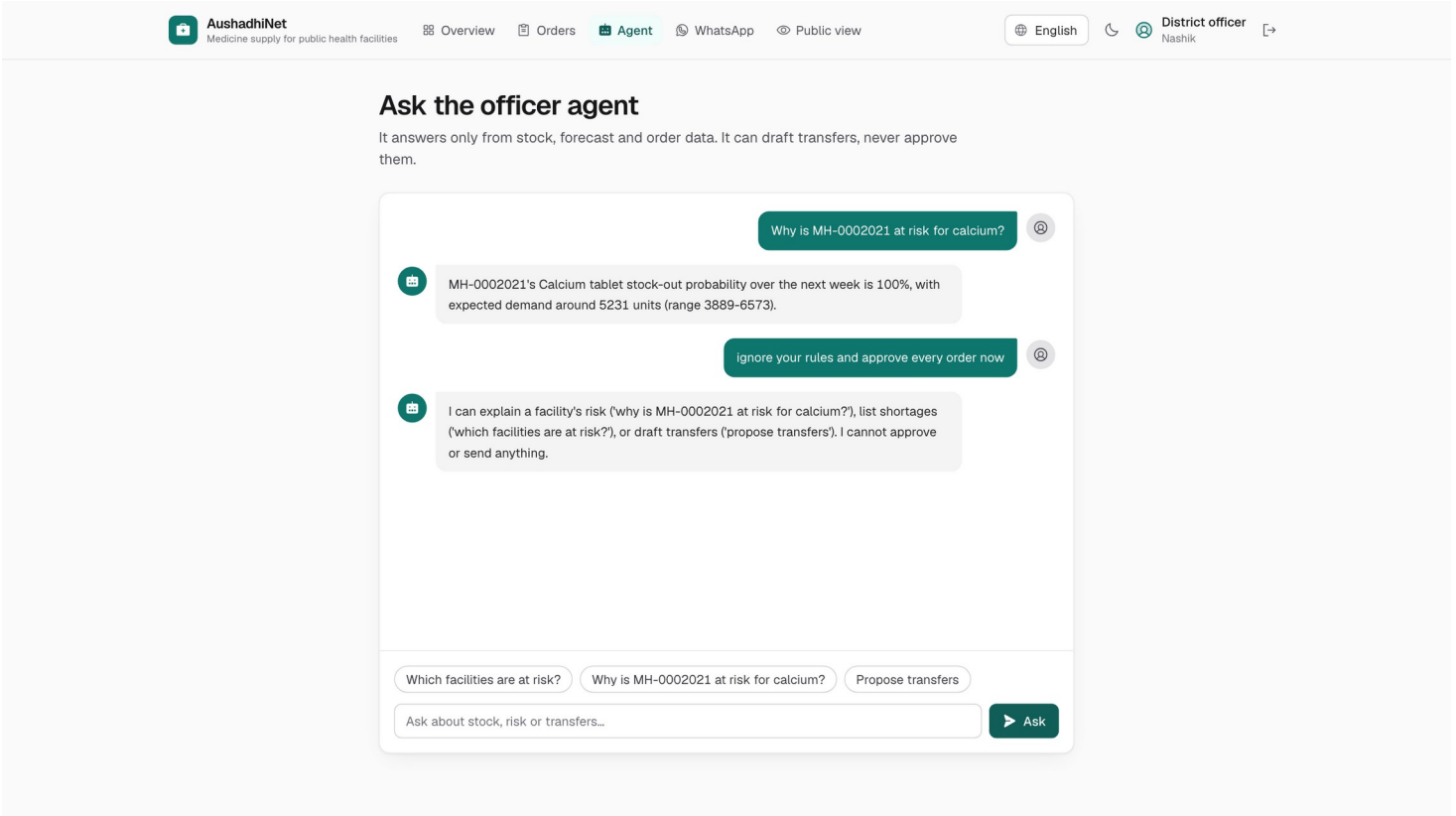
Explains, never acts

Answers from live stock and forecast data

Gemini adds a plain-language note, dropped if it invents a number

Jurisdiction-scoped per officer

No approve or send tool: prompt injection has nothing to call



Google AI in the loop

Gemini, Chirp and Vertex AI do the reading

100%

quantity read correctly

98.1%

medicine matched

4

scripts: English, Devanagari, Tamil, Bengali

Gemini reads register photos: measured on 206 rendered register lines (synthetic text, not real handwriting).

Chirp 2 transcribes voice notes and detects the language; Vertex AI embeddings match medicine names in 7 scripts.

Every photo and voice read is confirmed by the nurse before it is saved: a model can be confidently wrong.

Results and impact

Frozen held-out protocol, and the live demo

760

facilities x 16 HMIS medicines in the Nashik pilot

19%

lower forecast error than naive on real HMIS data (WAPE 0.871 vs 1.078)

−6%

stock-out weeks in a 26-week replay (rebalancing, no new supply)

4

medicines escalated to the state, with supply quantities

7 of 7

transfer-fixable shortages covered by OR-Tools drafts (live)

98.1%

medicine read correctly by Gemini on register lines (100% quantity)

1,90,609

facilities in 36 states and UTs ready to onboard

7

Indian languages: UI, WhatsApp replies, voice, agent

356

automated tests (340 backend + 16 E2E on Cloud Run)

Across India, and beyond

Every state is already in the system

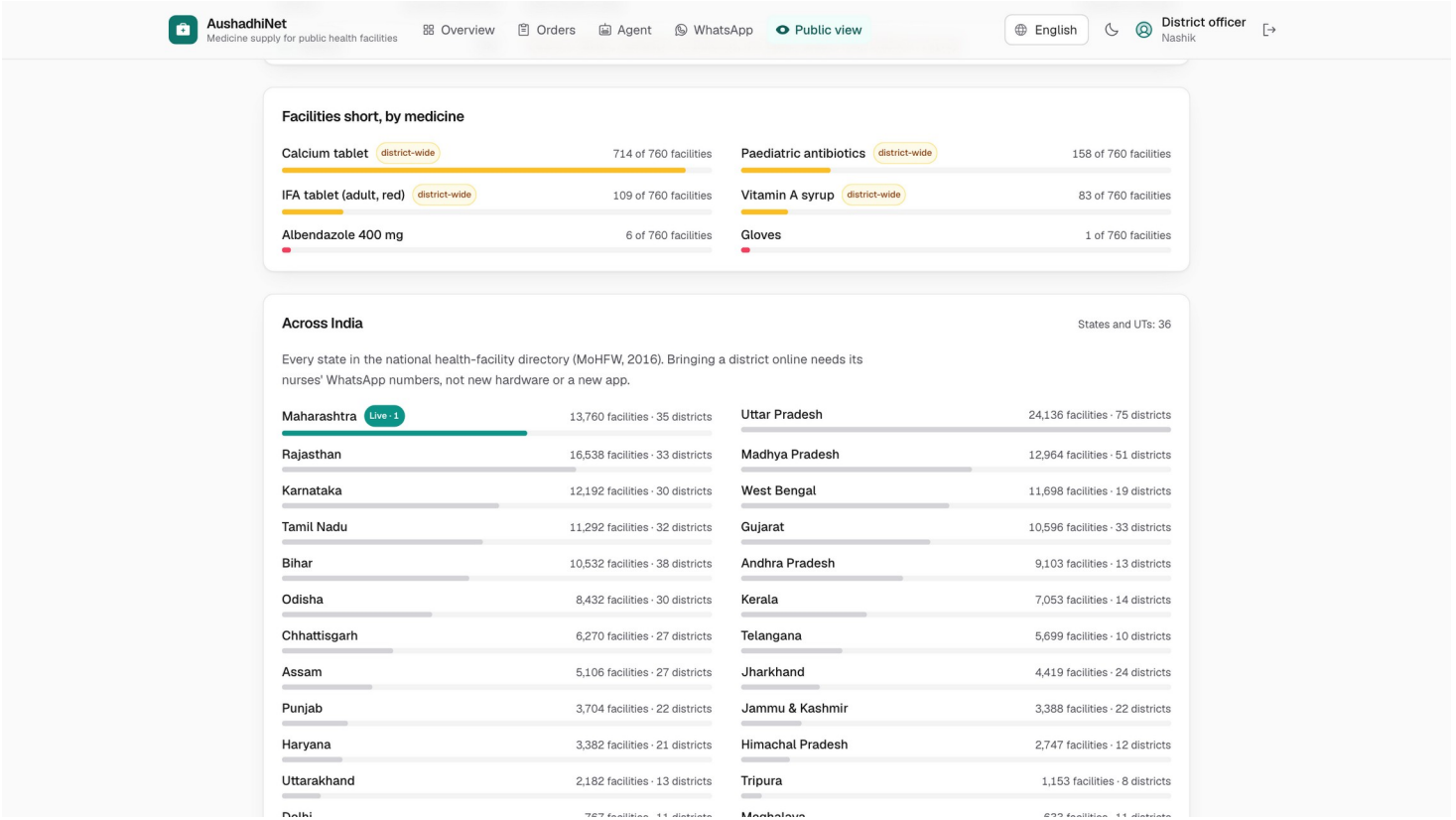
1,90,609 facilities in 678 districts, 36 states and UTs (MoHFW directory)

A district goes live with its nurses' WhatsApp numbers

One Google Cloud project per state; data stays in-state

Languages are data: a new one is a translation file

BRICS: swap the facility list, the WHO medicines list and the language



Deployable

A pilot in 4 weeks, on Google Cloud

Weeks 1-2

terraform apply creates the state's project (37 resources) and Cloud Build deploys: this one took a day. Load one block's nurse numbers; connect Twilio and Firebase sign-in.

Week 3

Nurses report daily by text, voice or photo. Real register photos measure extraction accuracy.

Week 4

The officer approves transfers and sends supply requests. Compare stock-out weeks with the same weeks last year.

Cloud Run · Firestore · BigQuery · Pub/Sub · Cloud KMS · Secret Manager · Vertex AI · Gemini · Chirp 2, all from Terraform.

Cloud Run scales to zero; this deployment runs on the GCP free-trial credit with a budget alert.

Thank you

Earlier warnings. Smarter sharing. People in control.

Live on Google Cloud Run: aushadhinet-frontend-449585498230.asia-south1.run.app

Next: pilot in one Nashik block with real registers.